

User Authentication

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Information Security

- Concerned with the:

★ identification of the information data assets

★ development and implementation of policies, standards, guidelines, and procedures

★ management practices of data classification and risk management

★ addressing confidentiality, integrity, and availability

★ identifying threats, classifying the organizations assets

★ rating assets vulnerabilities so that effective security controls can be implemented.

Security Concerns

- Three tenets: Confidentiality, Integrity, and Availability (C.I.A.), expanded into list of critical characteristics of information
- Identification, authentication, accountability, authorization, and privacy
- Reducing the impact of threats and the likelihood of their occurrence

THE BIG THREE

- Confidentiality

Prevent:
- intentional
- unintentional

unauthorized disclosure of contents.

- Integrity

- No modification to data by unauthorized person or Processes
- Data is consistent;

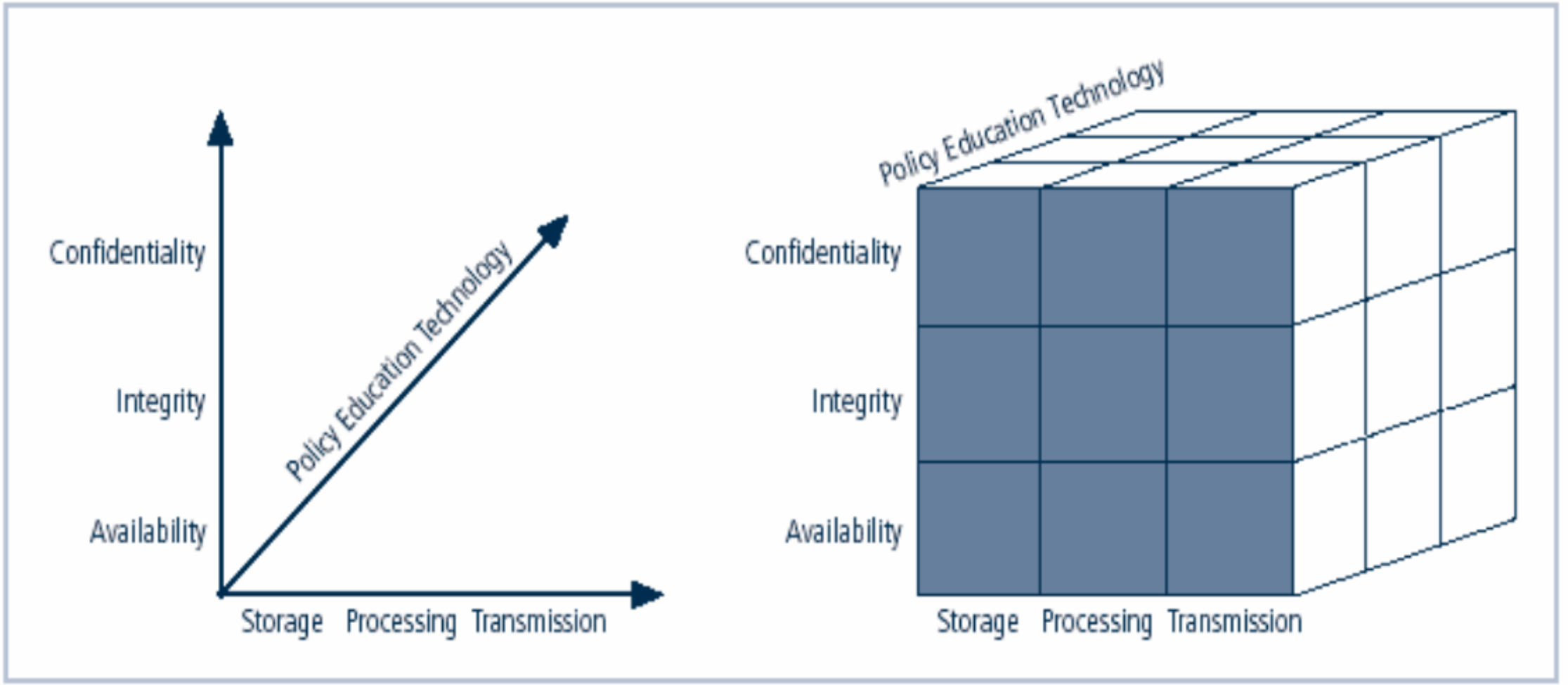
- Availability

Reliable and timely access to resources.

Critical Characteristics of Information

- The value of information comes from the characteristics it possesses:
  - Availability
  - Accuracy
  - Authenticity
  - Confidentiality
  - Integrity
  - Utility
  - Possession

NSTISSC Security Model



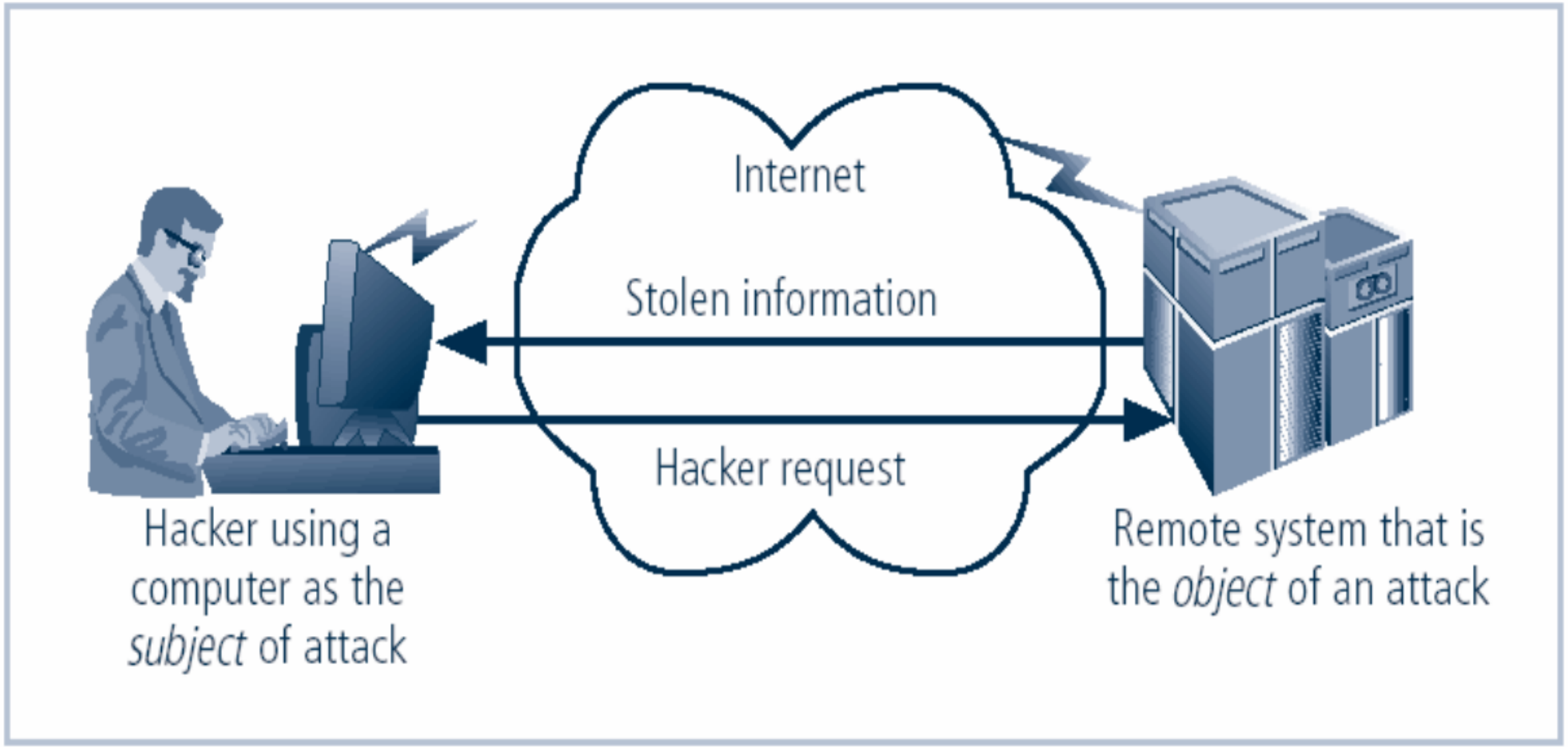
NSTISSC Security Model

Components of an Information System

- Information System (IS) is entire set of
  - ★Software
  - ★Hardware
  - ★Data
  - ★people
  - ★Procedures
  - ★networks

Securing Components

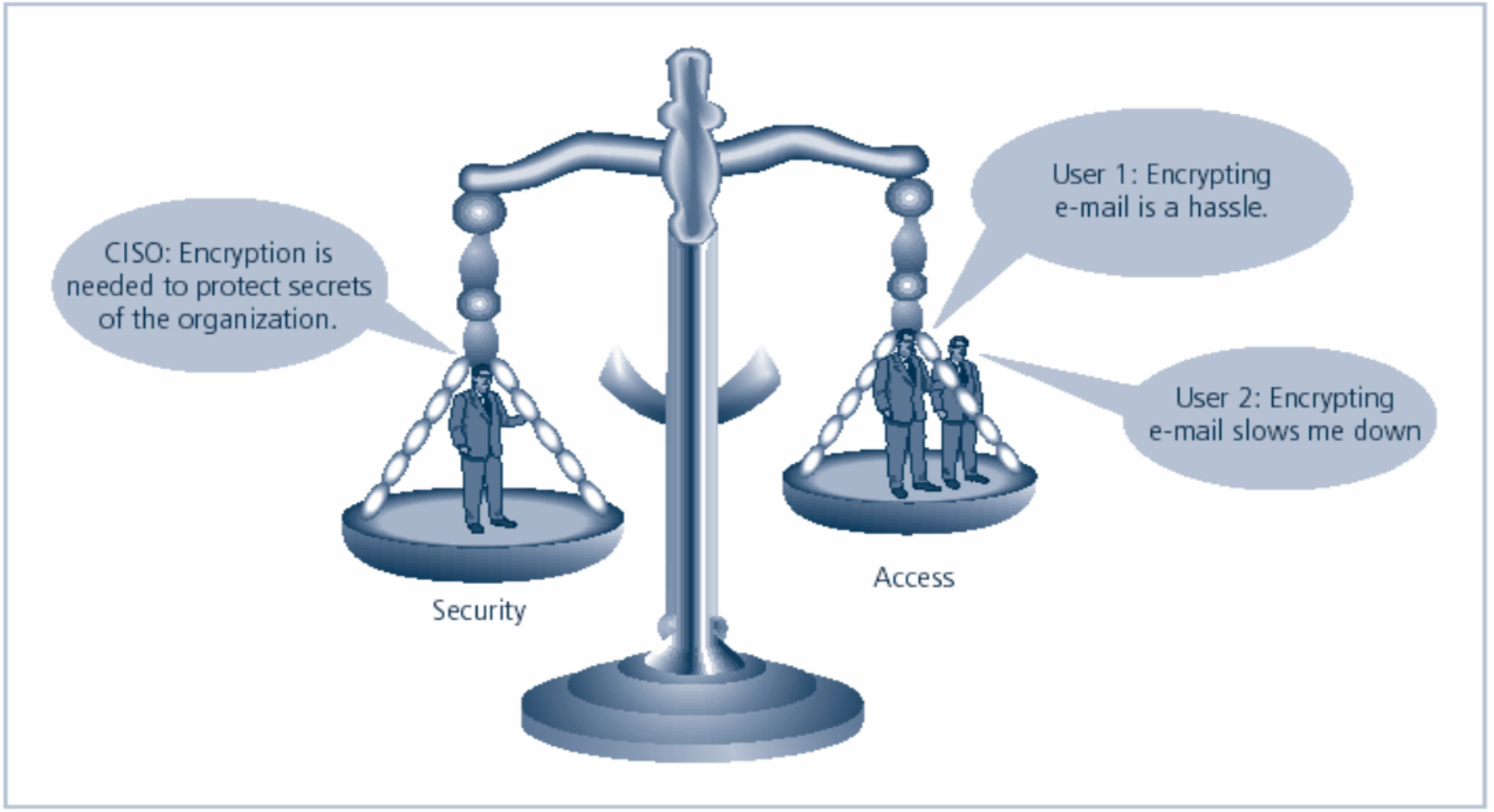
- Computer can be subject of an attack and/or the object of an attack
  - When the subject of an attack, computer is used as an active tool to conduct attack
  - When the object of an attack, computer is the entity being attacked



Computer as the Subject and Object of an Attack

Balancing Information Security and Access

- Impossible to obtain perfect security—it is a process, not an absolute
- Security should be considered balance between protection and availability
- To achieve balance, level of security must allow reasonable access, yet protect against threats



Balancing Information Security and Access

Objectives of Information Security

- Reduce the effects of security threats and vulnerabilities to tolerable levels
- To ensure correct
  - ★ Identification
  - ★ Authentication
  - ★ Accountability
  - ★ Authorization
  - ★ Privacy

- **Identification:** For access control users have to claim their identities to a system for authentication and authorization.
- **Authentication:** The evidence provided by a user is reconciled with shored details to establishes the truthiness of the details provided.
- **Accountability:** A system the actions and behaviours of an individual within a system to determine their identity using the audit trails and logs.
- **Authorization:** The levels which determine the rights and permissions of the extent of an individual is enabled to access data computer resource.
- **Privacy: C**onfidentiality and privacy given to a user in a system to guarantee the fundamental tenet of confidentiality of a data.

Organization Assets and Data Classification

- **Unclassified:** Information that does not violate confidentiality, designated as not sensitive
- **Sensitive but Unclassified (SBU):** Information with minor secret and will not cause serious damage if disclosed.
- **Confidential:** Information if unauthorizedly disclosed, can cause some damage to an organisation.
- **Secret:** Information if unauthorized disclosed, will cause serious damage to an organisation.
- **Top Secret:** The highest level of information classification, which by if unauthorizedly disclosed, will cause exceptionally grave damage to an organisation.

Legal Distribution and Discloser Classified Assets

- **Court order:** Disclosed to a judicial proceeding, to comply with a court order
- **Government contracts:** To disclose information in accordance with the procurement agreements that are related to a government project, required by LAW
- **Senior-level approval:** After signing a confidentiality none-discloser-agreement by the external party, a senior-level executive can authorize the release of information to facilitate a proceeding.

Key Terms

- Access
- Asset
- Attack
- Control, Safeguard or Countermeasure
- Exploit
- Exposure
- Hacking
- Object
- Risk
- Security Blueprint
- Security Model
- Security Posture or Security Profile
- Subject
- Threats
- Threat Agent
- Vulnerability